Pizza Sales Analysis

Project Overview

The Pizza Sales Analysis project was developed using MySQL and Microsoft Excel to analyze pizza sales performance, customer ordering behavior, and product trends.

The objective of this project is to transform raw transactional data into meaningful business insights through SQL analysis and interactive dashboard visualization.

Business Problem Statement

The business generates large amounts of pizza sales data daily, but without analysis it becomes difficult to understand:

- Customer ordering patterns
- Peak sales periods
- Best-selling products
- Revenue-driving categories
- Product performance trends

The goal of this project is to analyze sales data and create a dashboard that supports data-driven business decision-making.

Project Objectives

- Analyze total revenue and order performance
- Identify peak ordering days and hours
- Understand customer preference by pizza category and size
- Identify best-selling and worst-selling pizzas
- Build an interactive dashboard for business reporting
- Generate business recommendations from sales insights

Tools & Technologies Used

Tool	Purpose
MySQL	Data analysis and querying
Microsoft Excel	Dashboard creation
SQL	KPI and trend analysis
Pivot Tables & Charts	Data visualization
Slicers	Interactive filtering

Key Business Questions

- What is the total revenue generated from pizza sales?
- Which pizza categories contribute the highest sales?
- What are the peak ordering days and hours?
- Which pizza sizes are most preferred by customers?
- Which pizzas are best sellers and worst sellers?
- How can the business improve operational efficiency and sales performance?

Database Creation

A dedicated MySQL database named `Pizza_sales_analysis` was created to store and manage pizza sales data.

Table Structure

The `pizza_sales` table was created with the following important fields:

- pizza_id
- order_id
- pizza_name
- pizza_category
- pizza_size

- quantity
- total_price
- order_date
- order_time

The dataset was imported from a CSV file into MySQL using the Import Wizard.

Project Workflow

Step 1: Data Import

- Imported pizza sales CSV dataset into MySQL
- Verified records and column structure

Step 2: Data Cleaning

- Converted order_date into DATE format
- Validated imported data
- Checked sample records

Step 3: SQL Analysis

Performed KPI and trend analysis using SQL queries:

- Total Revenue
- Average Order Value
- Total Orders
- Total Pizzas Sold
- Daily & Hourly Trends
- Category-wise Analysis
- Best & Worst Sellers

Step 4: Dashboard Creation

Created an interactive Excel dashboard using:

- KPI Cards
- Bar Charts
- Pie Charts

- Line Charts
- Slicers

Step 5: Business Insight Generation

Generated business recommendations based on dashboard analysis.

Dashboard Insights & Business Recommendations

1. Peak Sales Timing

Insight

- Highest order activity occurs on Friday and Saturday
- Peak ordering hours are between 12 PM–1 PM and 5 PM–8 PM

Business Recommendation

Increase staffing, kitchen preparation, and inventory availability during peak hours to improve customer service and reduce delays.

---

2. Category Performance

Insight

- The Classic category contributes the highest share of total sales and orders.

Business Recommendation

Focus promotional campaigns and combo offers on high-performing categories to maximize revenue generation.

---

3. Pizza Size Analysis

Insight

- Large pizzas generate the highest sales contribution among all pizza sizes.

Business Recommendation

Introduce family meal deals and upselling strategies to increase average order value.

---

4. Best-Selling Products

Insight

Top-performing pizzas include:

- Pepperoni Pizza
- Barbecue Chicken Pizza
- California Chicken Pizza

These pizzas consistently attract high customer demand.

Business Recommendation

Feature best-selling pizzas in advertisements, banners, and combo offers to further boost sales.

---

5. Worst-Selling Products

Insight

Lowest-performing pizzas include:

- Brie Carre Pizza
- Calabrese Pizza
- Mediterranean Pizza

These products contribute the least to total sales.

Business Recommendation

Review pricing strategy, improve menu visibility, or redesign promotional campaigns for underperforming products.

SQL Concepts Used

- CREATE DATABASE
- CREATE TABLE
- UPDATE
- ALTER TABLE
- STR_TO_DATE()
- SUM()
- COUNT()
- DISTINCT
- ROUND()
- GROUP BY
- ORDER BY
- WHERE Clause
- Aggregate Functions
- Date Functions

Conclusion

This project demonstrates how SQL and Excel can be used together for business analysis and dashboard reporting.

The analysis helped identify customer behavior patterns, peak business periods, high-performing products, and revenue-driving categories.

The final dashboard provides a clear and interactive view of business performance, enabling better operational and marketing decisions.


PIZZA SALE
Total Revenue $69,793
Avg Order value $37.83
Total Pizzas Sold 4232
Total Orders 1845
Avg Pizzas Per Order 2.29
Peak Days and Hours
DAYS
Order activity rises sharply toward the weekend, especially Fri-Sat
HOURS
Peaks occur aroung 12-01 PM and 5-8 PM
Daily Trend for Total Orders
Sunday 199
Monday 220
Tuesday 242
Wednesday 222
Thursday 329
Friday 330
Saturday 303
Hourly Trend for Total Orders
11 100
12 206
13 201
14 166
15 132
16 165
17 193
18 208
19 177
20 148
21 96
22 52
23 1
Sales by Category & Size
CATEGORY
Classic category leads in both sales and total orders
SIZE
Large Size Pizza generate the highest sales & contribution
% of Sales by Pizza Category
24.44%
23.20%
25.69%
26.68%
Chicken
Classic
Supreme
Veggie
% of Sales by Pizza Size
30.01%
21.64%
1.83%
0.10%
46.42%
Large
Medium
Regular
X-Large
XX-Large
Total Pizzas Sold by Pizza Category
Classic 1257
Supreme 1044
Veggie 1018
Chicken 913
Best & Worst Sellers
BEST
Classic deluxe & chicken pizzas are the sellers and revenue generators
WORST
The Brie Carre Pizza is at bottom in both orders and revenue
Top 5 Best Sellers by Total Pizzas Sold
The Pepperoni Pizza 239
The Barbecue Chicken Pizza 211
The California Chicken Pizza 202
The Thai Chicken Pizza 199
The Sicilian Pizza 191
Bottom 5 Worst Sellers by Total Pizzas Sold
The Spinach Pesto Pizza 77
The Chicken Pesto Pizza 77
The Green Garden Pizza 75
The Mediterranean Pizza 68
The Calabrese Pizza 67
The Brie Carre Pizza 35
order_date
Jan 2015
MONTHS
JAN FEB MAR APR MAY JUN